

DESMOS CALCULATOR INSTRUCTIONS

CORE SKILLS (VIDEO PART 3)

Topic	Instructions	Notes
Convert Fraction -> Decimal Decimal -> Fraction Mixed Number -> Fraction Reduce Fractions	Press the circle button with fraction inside after typing number	-When the button is highlighted (white), it will return a fraction, otherwise decimal -To type a mixed number, type the whole number and numerator of fraction, then highlight the numerator & press “/”
Percentages	Type the number with % sign	-Can do calculations/solve equations with %’s by typing the % sign
Define a Number as a Letter to Carry It Over to Future Calculations in Exact Form	Type A = calculation formula	-This will store the exact decimal value (no rounding or truncating) in the specified letter to use it for calculations -Also allows you to just type the letter instead of having to type out the entire number with 10+ decimal places
Add a Slider	Type any letter other than x or y and click on the letter next to “Add slider” or Type letter = #	-You can click the value to change it or click on either interval value to end new interval values -The step value is how much the slider will move on each drag to the left/right
Distance Between 2 Points (i.e. Distance Formula)	Type distance((2,6),(-1,7) or distance(A,B) if A=(2,6) B=(-1,7)	-Returns the distance between the two defined points using the distance formula -Useful to find distance between endpoints of a diameter for a circle
Midpoint Between 2 Points	Type midpoint((2,6),(-1,7) or or midpoint(A,B) if A=(2,6) B=(-1,7)	-Returns the midpoint between the two defined points -Click “Label” to show the coordinates -Useful to find center of a circle if you have 2 endpoints of a diameter
Find the Mean/Median of a Data Set	Type mean(1,2,...) or median(1,2,...)	-Mean & median are accessed from the “FUNC” menu in the keyboard or you can type the words
Identify the # of Solutions to a Linear Equation	Graph $y =$ left side of the equation and $y =$ right side of the equation	-2 parallel lines = No Solution -1 line = Infinite Solutions -Intersecting lines = 1 solution
Solve a System of Equations by Graphing/Identify # of Solutions	Graph both equations and look for the intersection point(s) or parallel/overlapping lines	-2 parallel lines = No Solution -1 line = Infinite Solutions -Intersecting lines = 1 solution
Identify Table of Values as Solution to Given Graph or Function or Inequality or System of Inequalities	Graph the points & the function/inequality(ies) to see if they are on the function or in the solution set for inequality(ies)	-If given graph of function/inequality, first find the function/inequality (using regression, etc.), then graph the points

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Linear Regression or Find Slope/Y-int/Equation of a Line Given 2 Points or Point & Slope or Graph of Line	Create a table and enter the x values under x_1 and the y values under y_1 1) Type $y_1 \sim mx_1 + b$ or 2) Click Regression Button 	-Must use the $\sim$ and not = -Slope (m) & Y-intercept (b) are found -Note: If you have 1 point & slope, you must type in $y_1 \sim mx_1 + b$; you cannot use regression button which needs 2 points
Evaluate a Function/ Combination of Functions or Find the Y Value of a Function for a Given X Value	Enter function in form of $f(x)=$ <u>Method 1</u> Create a new entry and type in $f(\#)$ for the value to evaluate at $x = \#$ <u>Method 2</u> Click the Gear icon and chose Create Table icon; then type in the x-values you want to evaluate <u>Method 3</u> Graph the line $x = \#$ and then look for the intersection point	-Function must be $f(x) = \dots$ -Easier to use sometimes than tracing along the graph -For Method 2, type in the x-values you wish to evaluate and the $f(x)$ column will automatically update -Method 3 does not require you to type the function, i.e. if you have a graph from a regression, you can do Method 3 without retyping the full function -Method 3 can require a lot of scrolling if the x or y values are large -For combination of functions, type in each function and then do the operation, ex. $f(2) - g(1)$
Find X Value of a Function for a Given Y Value	Type in the function and graph $y = \#$ for the given value & look for intersection point(s)	-Easier to use sometimes than tracing along the graph
Solve an Equation for x *Linear *Absolute Value *Radical *Rational *Quadratic	<u>Method 1</u> Graph $y =$ left side of the equation and $y =$ right side of the equation; look for the intersection point(s), i.e. solutions <u>Method 2</u> Type in the equation as written and look for the vertical line(s)	Method 1: -Preferred because it will show you No Solution (no intersection) or Inf Solutions (same graph on top of itself); in Method 2, the screen is blank for both cases, so you don't know which one it is Method 2: -X-int(s) of vertical line(s) are solution(s) -Some equations (Radical, Abs Value, etc.) will not return all solutions -Some lines will not be clickable; to get the value do a regression (see below)
Solve an Equation using Regression	Type in the equation as written but replace the = sign with $\sim$ and all x 's with x_1	-Will return one solution only -Use $\{x>0\}$ to get other solutions -Useful for Pythagorean Theorem or when answers are decimals or very large
Translate/Shift a Graph	Graph first function as $f(x) =$ Enter a new function as $g(x) =$ $f(x)$ and add the shift	-This will allow you to find points on the translated function but not the actual translated function itself

DESMOS CALCULATOR INSTRUCTIONS

INTERMEDIATE SKILLS (VIDEO PART 4)

Topic	Instructions	Notes
Identify Radius & Center of Circle Given Equation	Graph the Equation Find the top & bottom endpoints of a diameter and find distance between them or subtract y-values for diameter length; then divide by 2 for radius Center is midpoint (average) of the y-values for top & bottom diameter points and the same x-value for each point	-The circle equation can be in any form, it does not need to be in standard form -Make sure you are clicking on the top & bottom diameter endpoints, NOT the y-intercepts which are sometimes very close to each other distance((a, b),(c, d)) gives distance between points (a, b) and (c, d) midpoint((a,b),(c,d)) will plot the midpoint of (a, b) and (c, d)
Find the Mean of a Data Set from a Frequency Table	Create a table; enter the values in column x_1 & frequency in column y_1 Create a new entry with $\frac{total(x_1 * y_1)}{total(y_1)}$	-The <i>total</i> function adds up all the values for each entry for $x_1 * y_1$
Quadratic Function Regression	Create a table; enter x values under x_1 & y values under y_1 <u>Standard Form</u> 1) Type $y_1 \sim ax_1^2 + bx_1 + c$ or 2) Click Regression Button  <u>Vertex Form</u> Type $y_1 \sim a(x_1 - h)^2 + k$	-Must use the $\sim$ and not $=$ -Standard Form regression requires at least 3 points -Standard Form can use only 2 points *IF* a or b or c are already known -Vertex Form only requires 2 points: 1 point + vertex; <u>you must enter the h/k values in the regression line and must type it manually (cannot use the Regression button)</u>
Identify # of Real Solutions to Quadratic Equation	Graph the function and look for # of x-intercepts	-The # of x-intercepts = the # of real solutions; add sliders if needed
Factor a Polynomial Expression	Graph the expression and look for the x-intercepts	-For an x-intercept value of "c", the polynomial will have a factor (x - c)
Identify Equivalent Expressions	<u>x and/or y in expression</u> Graph both expressions and look for overlap <u>No x or y in expression</u> Type original expression and look for same value in answers	-If the two expressions are identical, the graph will show both of them overlapping each other -If expressions have x & y, then use graph method; if they do not have x or y, add sliders for each letter and match up the identical numerical values

DESMOS CALCULATOR INSTRUCTIONS

ADVANCED SKILLS (VIDEO PART 5)

Topic	Instructions	Notes								
Solve a Formula for a Specific Letter	Method 1 One Letter = everything 1) Enter the formula and add sliders for every letter except the one on the left side (alone) 2) Type the answer choices (only the right side) 3) See which answer choice has the same number that you chose for the letter in the slider	Method 1 -Used when you have a formula that is given with one letter alone on one side (usually left) and every other term (number, letter, etc.) on the other side (usually right side)								
	Method 2 Multiple Letters on Both Sides 1) Type in the formula but replace the letter that you are solving for with y; this will graph a horizontal line 2) Add sliders for other letters 3) Type answer choices and see which one matches the original graph from step 1	Method 2 -Used when you have multiple terms on both sides of the equation. -Can replace the letter you are solving for with y or x and then use the answer choices to match that horizontal or vertical line								
	Methods 1 & 2-Alternate 1) Type formula as regression statement 2) Type answer choices and see which one matches the value for the letter in the regression statement	Method 1 & 2-Alternate -You can also pick values for 1 or more letters using sliders if a chosen value doesn't meet the given conditions in the problem or 2 answer choices have the same result								
Use Sliders in Table of Values for Regression	Add d (or any letter other than x or y) into the x ₁ column in a table of values and then create a slider	-Can be used for quadratics when you know the function has 2 x-intercepts and you know the vertex to find the relationship between a, b, c Ex. <table><tr><td>x₁</td><td>y₁</td></tr><tr><td>2</td><td>-6</td></tr><tr><td>2 + d</td><td>0</td></tr><tr><td>2 - d</td><td>0</td></tr></table> -Can be used for other functions when you know an x-int is above or below a certain value, i.e. (r, 0) where r > 5	x ₁	y ₁	2	-6	2 + d	0	2 - d	0
x ₁	y ₁									
2	-6									
2 + d	0									
2 - d	0									
Run a Regression on Specific Points Values	[f(7), f(-3)]~0 f(-9)~f(3)	-Sets the y-values of two points both equal to 0 (or whatever value is chosen) -Sets the y-values of two points equal without knowing what the y value is								

DESMOS CALCULATOR INSTRUCTIONS

ADVANCED SKILLS (VIDEO PART 5)

Topic	Instructions	Notes
Find Value of Constant(s) to Match Expressions Using Regression	Method 1: 1) Type the equation but replace all x's with x_1 and replace = with $\sim$ 2) Create a list for x_1 with at least 2 values $x_1=[1,2]$ Method 2: 1) Graph the side of the equation which does not have the constants (i.e. just x's) 2) Choose several points on the graph & type them into a table 3) Run a regression on those points using the other side of the equation	Method 1 -The regression will return the values of all coefficients (i.e. constants), letters other than x -You must create a list for x_1 with more than one entry; otherwise, it treats x_1 as a constant (not a variable) and does not return the correct answers Method 2 -Ideally you want to pick points that have integer values for both x & y-coordinates or for at least one of the coordinates -Otherwise, pick points that have decimal values that can easily be changed into fractions or that terminate
Find Regressions for Other Types of Functions From a Set of Points *Exponential *Rational *Radical *Circle	Create a table; enter x values under x_1 & y values under y_1 <u>Exponential</u> Type $y_1 \sim ab^{x_1}$ <u>Rational</u> Type $y_1 \sim \frac{a}{mx_1 + b}$ <u>Radical</u> $y_1 \sim a\sqrt{x_1 + b}$ <u>Circle</u> $x_1^2 + 4x_1 + y_1^2 - 8y_1 - 20 \sim (x_1 + 2)^2 + (y_1 - k)^2 - r^2$	-The # of points required = the # of constants in the function -The form of these types of functions could vary based on the given information in the problem; these are just general examples of exponential, rational, and radical functions -For circles, you must have each equation on one side = 0 and those two equations are then regressed to each other, one in standard form and the other in general form; must have same coefficients on the x_1 and y_1 terms on both sides and must make lists for both x_1 & y_1 (see above)
Use Sliders to Solve a Single Equation with a Constant	1) Graph $y =$ left side & $y =$ right side 2) Add a slider for constant(s) 3) Find the solutions (intersection points) on graph 4) Type in the answer choices to match the solution	-Can be difficult to find the intersection point(s) depending on the shape of the graphs; move the slider around as needed to make it easier to find
Solve Sys of Equations Using Regression	[equation 1 left side, equation 2 left side]~[equation 1 right side, equation 2 right side]	-Must put all variables as x_1, y_1, etc. -Returns the exact value and helpful when intersection point is difficult to find -Use $\{x_1 > 0\}$, $\{x_1 < 0\}$, etc. to find additional answers if needed

DESMOS CALCULATOR INSTRUCTIONS

ADVANCED SKILLS (VIDEO PART 9)

Topic	Instructions	Notes
Find Values of a Constant in Quadratic Solution or Equation	Method 1: 1) Type the equation as regression statement, i.e. replace all x's $\rightarrow x_1$ and $= \rightarrow \sim$ 2) Type $x_1 =$ expression you are given in the problem, ex. $x_1 = \frac{(4 + \sqrt{k})}{2}$ Method 2: 1) Type the equation as $f(x) =$ 2) Type $f(\text{expression you are given}) \sim 0$, ex. $f\left(\frac{(4 + \sqrt{k})}{2}\right) \sim 0$	Method 1 -The regression will return the value of the coefficient in the expression in the first statement (i.e. the $\sim$ statement) Method 2 -The second statement (regression) will return the value of the coefficient in the expression -The constant can also be in the equation itself; either method will work
Linear Regression Using Lists	1) Type $x_1 = [0, k - 2, k + 4]$ 2) Type $y_1 = [10, 20, 16]$ 3) Type $y_1 \sim mx_1 + b$	-The regression will return the values for m (slope), b (y-int), and k (constant) -You must define the point values in a list (not a table) for regression to work when there are constants and you are not using a slider for it
Find Value of a Constant for System of Equations to Have No Solution or Inf Solutions	1) Re-write both equations (manually) so that all x & y terms are on the same side 2) $\left(\frac{x \text{ coefficients}}{x \text{ coefficients}}\right) \sim \left(\frac{y \text{ coefficients}}{y \text{ coefficients}}\right)$ Eqn 1 coefficients = Top Eqn 2 coefficients = Bottom	-Both equations must have *ALL* x & y terms on the same side & both equations do not need to be on same side -You can type multiple coefficients for x and/or y -You can use this to find the ratio of multiple constants but the individual values *may not* be correct
Find Value of a Constant in a Sum/Product of Roots Problem	1) Given form of the equation $\sim$ intercept (factored) form 2) Define the roots using the given info about the sum/product of roots ex. Sum of roots = 15, so $a + b + c = 15$ Type $a = 15 - b - c$ 3) Create a list for x_1 with at least 3+ values $x_1 = [-2, 0, 2, 3]$	-The intercept (factored) form will look like $m(x_1 - a)(x_1 - b)(x_1 - c)$, etc. where a, b, c, are the roots of the equation -The Sum of Roots or Product of Roots will be given, and that equation must be re-arranged to be defined in terms of one of the letters -Will return the values of any constant(s) in addition to the roots themselves -Use negative #, 0, positive # in the list

DESMOS CALCULATOR INSTRUCTIONS

ADVANCED SKILLS (VIDEO PART 10)

Topic	Instructions	Notes
Find Value of Constant in Quadratic Expression for One Solution	1) Type the original quadratic expression with x_1 (make sure $= 0$) 2) Type $\sim$ 3) Type $a(x_1 - h)^2$ 4) Create a list for x_1 with at least 3+ values $x_1 = [-2, 0, 2, 3]$	-The original quadratic expression must be equal to 0 so move everything to one side if needed -Regression to the vertex form with a vertex of $(h, 0)$ forces one solution since vertex will be on the x-axis -Note: This can be used in non-quadratic equations as well if they can be turned into a quadratic by squaring both sides and then following the same general procedure
Find Value of a Constant for Quadratic & Linear Function to Intersect Once	Method 1: 1) Set the two functions equal to each other and then get the new function $= 0$ 2) Type the new function with x_1 2) Type $\sim$ 3) Type $a(x_1 - h)^2$ 4) Create a list for x_1 with at least 3+ values $x_1 = [-2, 0, 2, 3]$ Method 2: 1) Type in each function as $f(x)$ & $g(x)$ 2) Type $[f(x_1), f'(x_1)] \sim [g(x_1), g'(x_1)]$	Method 1: -Set the two functions equal (assuming they are both $y =$) and then move terms to one side to get everything $= 0$ -Regression to the vertex form with a vertex of $(h, 0)$ forces one solution (intersection) since vertex will be on the x-axis Method 2: 1) Solves a sys of equations where the functions themselves are equal (y-coordinates) and the derivatives (slope of the tangent line) are equal as well, both of which are true at the intersection point 2) Must use x_1 in regression statement 3) Displays the graph of both functions
Find Value of 1 or 2 Constant(s) for System of Equations to Have Inf Solutions (Alternate Method)	1) Move all terms to one side in both equations so that they are both $y =$ 2) Type Equation 1 $\sim$ Equation 2 (right side only for both with x_1) 3) Create a list for x_1 with at least 3+ values $x_1 = [-2, 0, 2, 3]$ 4) Type $\frac{c}{a}, \frac{r}{p}$, etc.	-Both equations must be $y =$ but you do not have to simplify the various terms. Ex. $y = \frac{\frac{2}{3}x - \frac{5}{2} + \frac{7}{3}x}{p}$ -You can have multiple terms for x and/or numbers

DESMOS CALCULATOR INSTRUCTIONS

ADVANCED SKILLS (VIDEO PART 11)

Topic	Instructions	Notes
Use Sliders to Solve a Single Equation with a Constant (Non-Graphing Method)	1) Type the equation as a regression statement using x_1 2) Add a slider for constant(s) 3) Regression will return a solution for x_1 4) Type in the answer choices to match the solution value	-Easier than finding the intersection point(s) on a graph -There may be multiple solutions; use condition of $\{x_1 > 0\}$ or $\{x_1 < 0\}$ in the regression statement to find a positive or negative solution if needed
Solve Advanced % Problems Using Regression	Method 1: 1) Type each equation as given in the problem 2) Last equation should be a regression statement with the % constant you are solving for, i.e p, q, n, etc. Method 2: 1) Create a system of equations to solve using regression 2) Type [Eqn 1 left, Eqn 2 left, Eqn 3 left] ~ [Eqn 1 right, Eqn 2 right, Eqn 3 right]	Method 1: -Can use % sign in the equations -The 2 equations must be equal to different letters (constants); otherwise, it will not work, i.e. you cannot have 2 equations that are both a = ... Thus, you may need to rewrite one of the equations in terms of a different letter Method 2: -You do not need to rewrite/change any of the equations for this method -Can use % sign in the sys of equations
Find Value of a Constant(s) for System of Equations to Have No Solution/Infinite Solutions (Alternate Methods)	Method 1: 1) Re-write both equations (manually) so that all x & y terms are on the same side 2) Using x_1 and y_1 , type [Eqn 1 left, Eqn 2 left] ~ [0,0] 3) The regression will return values for the constant(s) Method 2: 1) Type [Eqn 1 left, Eqn 2 left] ~ [Eqn 1 right, Eqn 2 right] using x_1 and y_1 but remove, i.e. do not type, the constant terms 2) The regression will return the values for the constant(s)	Method 1: -You do not have to combine/simplify multiple terms of x and/or y -The x & y terms can be on either side, but must be separated from the # term Method 2: -You do not need to move any terms Both Methods: -Works for all No Solution problems & Inf Solution problems where the constant(s) are on x and/or y terms only, i.e. cannot solve of constant of # terms **Will need constraint $\{x_1 > 0\}$ or $\{x_1 > 1\}$ when constant is in denominator**
Max/Min Value for a Constant in Quadratic Equation to Have No Real Solutions (Derivative Method)	1) Type $f(x)$ = quadratic function 2) Type $g(x) = 0$ 3) Type $[f(x_1), f'(x_1)] \sim [g(x_1), g'(x_1)]$	-The quadratic equation must = 0 -Value may be max or min depending on which constant (b or c) you solve for -b could have both max & min value while c has only max or min, not both -Check the opposite value or put in a condition $\{b > 0\}$ or $\{b < 0\}$ for other value